Task 12 Static Audit: Simulation Boundary Conditions and Scope

Implementation audit

2026-07-11

Contents

1 Audit rule and scope

This is a static audit of already-versioned simulation scripts and result CSVs. It does not run or source any simulation, create any Monte Carlo draw, or modify a raw/result artifact. The companion script, `quality_reports/build_task12_boundary_conditions_audit.R`, reads the Task 10 inventory and existing CSVs and writes only the audit products under `quality_reports/`.

The evidentiary labels have a deliberately narrow meaning:

- **Existing result** means that Task 10 records both a source script and a declared versioned result artifact, and the Task 12 builder confirmed that both paths exist.
- **Declared output missing** means that the source script specifies an output path but the audited tree contains no corresponding versioned result. It cannot support a numerical conclusion.
- **Legacy artifact** means that a retained exploratory script has no persisted result artifact. It is not numerical evidence.

This audit reports the magnitudes stored in the existing result CSVs only to delimit the represented DGPs. A coefficient shift becomes causal bias only with the manuscript's DAG, timing, target estimand, and identification assumptions.

2 Boundary-condition record

Table 1 records, family by family, what the existing code actually varies and holds fixed; what the retained artifacts measure; what remains unidentified; provenance; replication counts; and the appropriate interpretation boundary. A requested replication count is not silently substituted for the count actually persisted in the summary CSV when unstable draws were discarded.

Table 1: Static boundary-condition record for the existing simulation artifacts.

Family	Design: varied versus fixed	Evidence, magnitude, and replications	Not identified and interpretation boundary	Task 10 inventory IDs
Panel length, N, and burn-in	Varied: In the retained eight-model grid, T = 10, 20, 30, 50, 100 and rho_Z = 0.50, 0.85. Fixed: The retained T grid fixes N = 100 and burn-in = 100; it is not an N or burn-in sensitivity analysis.	Status: existing result; plus declared output missing. Bias and MCSE by T for eight estimators; |bias(ADL+FE+lagged Z)| = 0.003 to 0.014. Replications: 500 per retained scenario	No result identifies robustness to N or burn-in changes, arbitrary short panels, or a general finite-T correction. The separate 200-rep T script declares an output but has no versioned CSV. Scope: Figure 2's T = 30 result is conditional on its fixed N, burn-in, DGP, and estimator set; it is not evidence that dynamic FE is generally free of finite-T bias.	dual_role_varyT_8models; dual_role_varyT (declared output missing)
Binary and staggered absorbing treatment	Varied: Switcher/never-treated shares, onset times, collider channels, outcome persistence, and rho_Z in binary dynamic and absorbing-staggered DGPs. Fixed: Dynamic binary designs retain a linear homogeneous CET (beta = 1), short panel windows, and their stated lag structure; treatment is exogenous in mechC_adl.	Status: existing result. Bias/RMSE/MCSE by specification. mechC_adl has 48 retained scenarios; absorbing design has 24. Replications: 500 per retained scenario in each cited dynamic result CSV	No result identifies effects for non-absorbing treatment paths, anticipation, treatment-effect heterogeneity, regime effects, or arbitrary adoption processes. Scope: These are design-specific CET comparisons under stated adoption clocks; they do not validate a general DID, event-study, or treatment-regime estimator.	mechC_adl; staggered_posttreat
Nonlinear treatment and covariate response	Varied: Specified quadratic, cubic, logarithmic, sinusoidal, interaction, nonlinear carryover, and level-dependent-trend forms and strengths. Fixed: Most nonlinear designs fix N = 100, T = 30, burn-in = 100, linear CET beta = 1, and the remaining dynamic parameters; the collider grid also includes T = 10, 30.	Status: existing result. Bias, RMSE, MCSE, and discarded draws. Persisted replications: collider (requested 500; persisted 76–500 per scenario), interaction (500 per retained scenario), carryover (requested 500; persisted 445–500 per scenario); stable Callaway grid max |ADL(all) bias| = 0.000 to 0.004. Replications: collider requested 500; persisted 76–500 per scenario ; interaction 500 per retained scenario ; carryover requested 500; persisted 445–500 per scenario	No result identifies arbitrary nonlinear links, untested interactions, nonlinear outcome models, or general causal effects outside the named DGPs. Scope: The results test only named stable functional forms and a linear CET target; they bound neither unknown nonlinearities nor link-function misspecification generally.	nl_collider; nl_interact; nl_carryover; callaway_controls
Measurement error in the observed control	Varied: Classical measurement-error variance in observed Z: sigma^2_me = 0, 0.5, 1, 2, with collider-channel strengths. Fixed: N = 200, T = 30, beta = 1, no burn-in, and a static TWFE collider DGP are fixed in the legacy v4 mechanism-D artifact.	Status: existing result. FWL shift, coefficient bias, RMSE, and coverage; |long-model bias| = 0.058 to 0.530. Replications: 500 per scenario (legacy v4 CSV)	No result identifies the latent true Z, corrects measurement error, or validates the specification-shift interpretation when measurement reliability is unknown. Scope: Measurement error changes the observed projection and attenuation pattern; it does not turn the observed shift into a causal estimate or supply reliability correction.	v4_mechD
Lagged feedback from outcome to treatment	Varied: Lagged feedback phi from Y_(t-1) to D_t and rho_Z; related direct-feedback and feedback-plus-carryover designs vary their documented grids. Fixed: N = 100, T = 30, burn-in = 100, beta = 1, and the stated dual-role channels are fixed; feedback is lagged, not contemporaneous.	Status: existing result. Bias, RMSE, MCSE, stability/discard records, and FWL quantities. Retained Y-to-D rows: 7; 500 per retained scenario. Replications: Y-to-D 500 per retained scenario ; direct-feedback 500 per retained scenario ; feedback-carryover 500 per retained scenario	No result covers contemporaneous Y_t-to-D_t feedback, all feedback lags, or a causal estimator without sequential exchangeability and sufficient observed history. Scope: Lagged feedback can be studied only within the stated stable grid; the conditional baseline still requires a defended causal clock, history, and exchangeability.	feedback_Y_to_D; direct_feedback; feedback_carryover
Carryover and lag specification	Varied: Linear beta_2, nonlinear carryover form/strength, and rho_Z; combined design also varies feedback phi. Fixed: N = 100, T = 30, burn-in = 100, beta = 1, and the stated dual-role channels are fixed in the direct/nonlinear designs.	Status: existing result. Bias, RMSE, MCSE, estimated lag coefficient, and stability/discard records. Direct carryover: 500 per retained scenario; feedback-carryover: 500 per retained scenario. Replications: direct 500 per retained scenario ; nonlinear requested 500; persisted 445–500 per scenario ; feedback-carryover 500 per retained scenario	No result identifies cumulative/long-run effects when the lag structure is wrong, or establishes that a single-lag ADL is adequate in other DGPs. Scope: Correctly representing the relevant history is a scope condition. If the target is cumulative or the lag order is wrong, the reported coefficient can answer a different question.	direct_carryover; feedback_carryover; nl_carryover
Contemporaneous unobserved confounding	Varied: Current confounding strength kappa, persistence phi_U, rho_Z, and the U-to-Z channel delta_U. Fixed: N = 100, T = 30, burn-in = 100, beta = 1, and the specified linear dynamic structure are fixed.	Status: existing result. Bias/RMSE/MCSE for observed-history specifications and an oracle observing U_t; with kappa > 0, |ADL(all) bias| = 0.079 to 0.526 and oracle |bias| = 0.000 to 0.003. Replications: 500 per retained scenario	No observed-history specification identifies the CET with an omitted current common cause; the oracle is a DGP benchmark, not an empirical remedy. Scope: The observed-history ADL baseline does not solve contemporaneous unobserved confounding; an additional design, measurement, or assumption is required.	persistent_unobserved

3 Reader-facing compact table

The manuscript uses the following compact version after Figure 2. It is a map of what the reported simulations delimit, not a claim that the conditional baseline is generally robust.

Table 2: Boundary conditions represented by existing Monte Carlo artifacts. Each row is a design-specific result, not a general robustness or identification claim. Exact grids and provenance are in the Task 12 audit.

Boundary examined in existing DGPs	Existing evidence	Interpretation boundary
Panel length, N, and burn-in	T varies from 10 to 100 with N and burn-in fixed; the separately declared 200-rep T output is absent.	Figure 2's T = 30 result is conditional on its fixed N, burn-in, DGP, and estimator set; it is not evidence that dynamic FE is generally free of finite-T bias.
Binary and staggered absorbing treatment	Binary staggered and absorbing-treatment designs vary adoption composition and timing (500 retained replications per scenario).	These are design-specific CET comparisons under stated adoption clocks; they do not validate a general DID, event-study, or treatment-regime estimator.
Nonlinear treatment and covariate response	Named nonlinear collider, interaction, carryover, and trend designs report bias/RMSE only for their retained stable scenarios.	The results test only named stable functional forms and a linear CET target; they bound neither unknown nonlinearities nor link-function misspecification generally.
Measurement error in the observed control	A legacy static TWFE design varies measurement-error variance in Z (500 replications per scenario).	Measurement error changes the observed projection and attenuation pattern; it does not turn the observed shift into a causal estimate or supply reliability correction.
Lagged feedback from outcome to treatment	Lagged-feedback and feedback-plus-carryover designs report bias and stability within their stated grids.	Lagged feedback can be studied only within the stated stable grid; the conditional baseline still requires a defended causal clock, history, and exchangeability.
Carryover and lag specification	Linear and nonlinear lag structures are varied; omitted or misspecified history can change the estimand.	Correctly representing the relevant history is a scope condition. If the target is cumulative or the lag order is wrong, the reported coefficient can answer a different question.
Contemporaneous unobserved confounding	When a current common cause is unobserved, observed-history ADL retains material bias; the oracle observes that cause.	The observed-history ADL baseline does not solve contemporaneous unobserved confounding; an additional design, measurement, or assumption is required.

4 Reconciliation with the Task 10 inventory and manuscript

Table 3: Reconciliation of the Task 10 inventory with the current PA manuscript claims.

manuscript__or__inventory__claim	inventory__audit	disposition
Figure 2 reports N = 100, T = 30, burn-in = 100, and 500 replications.	Consistent: Task 11 validation records 500 raw/published replications for the Figure 2 source, and Task 10 verifies its source/results artifacts.	no correction required
The manuscript reports stable-scenario nonlinearity evidence and separately reports remaining bias under contemporaneous unobserved confounding.	Consistent: Task 10 classifies Callaway and persistent-unobserved artifacts as verified existing results; the manuscript conditions the nonlinear statement on stable scenarios and says the baseline does not address unobserved current confounding.	no correction required
The manuscript must not use sim__dual__role__z__varyT.R to support a numerical finite-T claim.	Confirmed discrepancy/gap: Task 10 classifies dual__role__varyT as declared__output__missing. Its script states 200 reps, but no matching versioned result exists.	exclude from numerical claims
Measurement-error and binary/staggered artifacts are existing evidence but do not currently support a general estimator claim in the manuscript.	Clarification added: the compact table labels these as design-specific existing artifacts, not evidence of general robustness or identification beyond their DGPs.	scope language added
Legacy sim__ivb__completa.R remains outside numerical manuscript support.	Confirmed: Task 10 classifies v1__legacy__completa as legacy__no__persisted__result; it is not cited as numerical evidence.	exclude from numerical claims

The material correction is negative rather than additive: no numerical finite-T conclusion can be drawn from `sim_dual_role_z_varyT.R`, because its declared 200-rep output is absent. The existing eight-model finite-T result is distinct, has a saved CSV, and keeps $N = 100$ and burn-in 100 fixed. Similarly, `sim_ivb_completa.R` remains a legacy exploratory source without a persisted result and is excluded from numerical support.

5 Validation record

Table 4: Static validation checks performed by the Task 12 builder.

check	value	pass
required Task 12 inventory rows present	14	TRUE
source scripts exist	14	TRUE
declared result artifacts for verified rows exist	35	TRUE
dual_role_varyT is explicitly classified as missing output	declared_output_missing	TRUE
v1_legacy_completa is explicitly classified as legacy without persisted result	legacy_no_persisted_result	TRUE
no simulation code sourced or executed by this script	static reads only	TRUE

The complete source-script paths and result-artifact paths are retained, without abbreviation, in `quality_reports/task12_bound` the concise Task 10 inventory identifiers in Table 1 keep the rendered audit legible. The protected simulation directory is checked again after rendering against the SHA-256 manifest created before this audit. Rendering this report reads only the generated Task 12 tables and does not execute a simulation.